

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE307	INTERNETWORKING ESSENTIALS	0	0	2	1	
Course Weightage	ATT: 5 CAP: 45 ETP: 50	Exam Category: X6: Mid Term Exam: Not Applicable – End Term Exam: Practical				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT,ENTREPRENEURSHIP					

Course Outcomes :Through this course students should be able to

CO1 :: Understand the concept of internetworking devices and associated protocols

CO2 :: Perform the working of various commands for internetworking

CO3 :: Employ routing protocol for static and dynamic routing using VLSM and FLSM

CO4 :: Demonstrate the practical utilization of server configuration

CO5 :: Apply the concepts of classful and classless IP addressing scheme

CO6 :: Establish the end to end connectivity using network utilities in a simulation environment

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	COMPUTER NETWORKS	ANDREW S. TANENBAUM	PEARSON

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	DATA COMMUNICATION AND NETWORKING	BEHROUZ A. FOROUZAN	MCGRAW HILL EDUCATION

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://www.packettracernetwork.com/labs/packettracerlabs.html	Packet Tracer Tutorial
RW-2	https://www.firewall.cx/networking/network-cabling/network-cabling-utp-xover.html	Straight and Cross Cable

Scheme for CA:

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Component	Weightage (%)	Mapped CO(s)
Recoup CA		
Practical Work 1	50	CO1, CO2, CO3
Practical Work 2	50	CO4, CO5, CO6

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Practical Work 1	The objective is to give students hands-on experience with networking devices, bridging theory and real-world applications to develop key networking skills.	This lab evaluation provides students with a hands-on opportunity to delve into the practical workings of networking devices. Through this experience, students will: • Explore the functionality of key devices: Understand how hubs, switches, and routers operate in a network and different topologies. • Gain practical skills: Learn how to configure and work with these devices to build. This approach bridges theoretical knowledge with real-world applications, empowering students to develop essential networking skills.	Individual	Offline	100	3 / 5
Practical Work 2	The objective is to provide an advanced hands-on experience in network design and management, focusing on server integration, subnetting, and mastering networking commands.	The second lab evaluation builds upon the foundation established in the first session, offering students a more advanced, hands-on experience in network design and management. Key objectives include: • Server Integration: Students will learn to configure and incorporate servers into the network, enhancing its functionality. • Subnetting Concepts: A deep dive into subnetting will enable students to efficiently design and segment networks for better performance and security. • Networking Commands: Practical exercises will help students master essential commands for monitoring. This lab evaluation not only reinforces the concepts from the first session but also equips students with advanced networking skills to handle real-world scenarios.	Individual	Offline	100	10 / 12
Recoup CA	subjective test based on the syllabus	Deactivated 10June 25Additional chance of CA	Individual	Online	30	8 / 10

MOOCs/ Certification etc. mapped with the Academic Task(s)

Academic Task	Name Of Certification/Online Course/Test/Competition mapped	Type	Offered By Organisation
Practical Work 1	JUNOS, ASSOCIATE JNCIA-JUNOS	Industry Certification	JUNIPER NETWORKS
Practical Work 1	SMART INDIA HACKATHON	Competition/Hackathon	MINISTRY OF EDUCATION
Practical Work 2	JUNOS, ASSOCIATE JNCIA-JUNOS	Industry Certification	JUNIPER NETWORKS
Practical Work 2	SMART INDIA HACKATHON	Competition/Hackathon	MINISTRY OF EDUCATION

- Where MOOCs/ Certification etc. are mapped with Academic Tasks:
1. Students have choice to appear for Academic Task or MOOCs etc.
 2. The student may appear for both, In this case best obtained marks will be considered.

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	Network hardware and basics of IP addressing concept	Familiar with the usage of cisco packet tracer	RW-1 RW-2	How to use the Cisco packet tracer
Practical 2	Network hardware and basics of IP addressing concept	Basic working of hub, switch and router		How various networking devices work, How to add various interfaces to the devices available in the Cisco packet tracer
	Network hardware and basics of IP addressing concept	Adding of interfaces (ports) in the devices		How various networking devices work, How to add various interfaces to the devices available in the Cisco packet tracer
Practical 3	Network hardware and basics of IP addressing concept	Creation of cross and straight cables		How to prepare the cross and straight cables as well as know about the basics of IP addressing
	Network hardware and basics of IP addressing concept	IP addressing basics		How to prepare the cross and straight cables as well as know about the basics of IP addressing
Practical 4	Network hardware and basics of IP addressing concept	Implementation and configuration of different topologies such as bus, star, ring, mesh, tree and hybrid		How to design the network by connecting various devices, and how to configure the various devices by using commands
	Network hardware and basics of IP addressing concept	Configuration of devices using command line interface (CLI)		How to design the network by connecting various devices, and how to configure the various devices by using commands
Practical 5	Network Commands	ipconfig, ping, tracert, arp, netstat, nslookup, switch commands, router commands (use DOS and scenario based commands)		How to analyze the workings of the whole network by using various commands
Practical 6	Network layer routing protocols	Implementation of static routing using classfull addressing		How to communicate different networks with each other by using the concept of static routing

Practical 7	Network layer routing protocols	Implementation of dynamic routing using classfull addressing by configuring routing information protocol (RIP)		How to communicate different networks with each other by using the concept of dynamic routing
Practical 8	Network layer routing protocols	Implementation of static routing using classless addressing (FLSM and VLSM)		How to communicate different networks with each other by using the concepts of static and dynamic routing with a fixed as well as variable subnet approach
	Network layer routing protocols	Implementation of dynamic routing using classless addressing (FLSM and VLSM) by configuring routing information protocol (RIP, RIP version 2 for VLSM)		How to communicate different networks with each other by using the concepts of static and dynamic routing with a fixed as well as variable subnet approach
Practical 9	Network layer routing protocols	Implementation of dynamic routing using classless addressing (FLSM and VLSM) by configuring routing information protocol (RIP, RIP version 2 for VLSM)		How to communicate different networks with each other by using the concepts of static and dynamic routing with a fixed as well as variable subnet approach
	Network layer routing protocols	Implementation of static routing using classless addressing (FLSM and VLSM)		How to communicate different networks with each other by using the concepts of static and dynamic routing with a fixed as well as variable subnet approach
Practical 10	Network layer routing protocols	Implementation of static routing using classless addressing (FLSM and VLSM)		How to communicate different networks with each other by using the concepts of static and dynamic routing with a fixed as well as variable subnet approach
	Network layer routing protocols	Implementation of dynamic routing using classless addressing (FLSM and VLSM) by configuring routing information protocol (RIP, RIP version 2 for VLSM)		How to communicate different networks with each other by using the concepts of static and dynamic routing with a fixed as well as variable subnet approach
Practical 11	Server Configuration and LAN Setup	Implementation of LAN with DHCP configuration		How multiple IP's can be allocated to different end-terminal devices automatically and how to configure different services like DNS, FTP, HTTP, and email in the network
	Server Configuration and LAN Setup	Implementation of DNS, FTP, HTTP and Email server configuration		How multiple IP's can be allocated to different end-terminal devices automatically and how to configure different services like DNS, FTP, HTTP, and email in the network
Practical 12	Server Configuration and LAN Setup	Implementation of DNS, FTP, HTTP and Email server configuration		How multiple IP's can be allocated to different end-terminal devices automatically and how to configure different services like DNS, FTP, HTTP, and email in the network
	Server Configuration and LAN Setup	Implementation of LAN with DHCP configuration		How multiple IP's can be allocated to different end-terminal devices automatically and how to configure different services like DNS, FTP, HTTP, and email in the network

Practical 13	IPv6 addressing and routing	Implementation of IPv6 static routing		How to communicate different networks with each other by using the concept of static routing with the IPv6 addressing concept
Practical 14	IPv6 addressing and routing	Implementation of IPv6 dynamic routing		How to communicate different networks with each other by using the concept of dynamic routing with the IPv6 addressing concept
	SPILL OVER			
Practical 15	Spill Over			